

THE DYNAMICS OF A MODEL OF A PLANKTON–NUTRIENT INTERACTION

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A stability analysis is given for a model of plankton dynamics introduced by Wroblewski *et al.* (*Global Biogeochem. Cycles* 2, 199–218, 1988). The detailed dependence of the steady-states and their stability on the various model parameters is explicitly presented and analysed. It is shown that under certain conditions the coexistence of phytoplankton and zooplankton occurs in an orbitally stable oscillatory mode. A distinguished parameter is varied and the steady-states computed. The significance of the lack of stable steady-states leading to periodic population levels is investigated and related to certain oceanographic data.

1. Introduction. In a recent article Wroblewski *et al.* (1988), introduced a two trophic level food chain model which they used for the study of plankton densities in different ocean layers. Their study was based on an analysis of the steady-states of their system and did not treat the stability of these steady-states or the question of the existence of oscillatory stable solutions. The purpose of the present paper is to fill this gap in the analysis of the model, and to pursue the biological implications of the stability results and of the existence of stable oscillatory states.

Mathematical models of increasing complexity that describe the dynamics of marine plankton have been constructed since the pioneering paper of Riley *et al.* (1949) where a simple diffusion model was proposed. A number of the later models were mainly concerned with phytoplankton–herbivore interactions. Steele (1974) and Steele and Frost (1977) treat such models. The importance of nutrients to the growth of plankton leads to explicit incorporation of nutrient concentrations in the plankton–herbivore models of Evans and Parslow (1985), Frost (1987), Taylor (1988) and Wroblewski *et al.* (1988), to which we

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refer for a review of this type of model. Models that analyse the spatial distribution of plankton have also been extensively studied, and the reader is referred to Levin (1980) and the references therein for an overview of these types of models. Here we shall not model the spatial distribution of plankton but only consider water layers or columns (ocean, lake, etc) with spatially uniform nutrient and light levels which are stable for the duration of the applicability of our model. Periodic or random perturbation can be introduced into the model that we study. However, we shall not pursue this here.

The model that we treat has three interacting components consisting of the phytoplankton (P), herbivorous zooplankton (Z) and dissolved nutrients (N). Following Wroblewski *et al.* (1988), the two plankton levels are modelled in terms of their nitrogen content, N , which is assumed to be the nutrient primarily responsible for limiting phytoplankton reproduction. The total nitrogen level of this trophic chain is assumed to be constant for the duration of the study. That is we are studying a homogeneous layer (or column) of water within which there is perfect mixing but there is no exchange with adjacent layers (or columns). Thus:

$$P + Z + N = N_T = \text{constant}, \quad (1)$$

where N_T like some of the other parameters that enter the model, may vary from one non-mixing layer (or column) to another. The pertinent parameters which enter the model are (our symbols differ from those of Wroblewski *et al.*):

- a maximal nutrient uptake rate for the phytoplankton
- k Michealis–Menton half saturation constant
- r phytoplankton mortality rate
- b maximal zooplankton ingestion rate
- c per capita saturation exponent
- δ zooplankton nutrient conversion rate
- ε zooplankton death rate
- δ_1 zooplankton nutrient loss rate.

The values of these parameters used by Wroblewski *et al.* are shown in Table 2.

The model assumes that there is no net nutrient loss due to death or due to nutrient conversion, hence, in particular:

$$b = \delta + \delta_1. \quad (2)$$

The resulting model equations (3) are given in the next section, and a detailed discussion of the model is found in Wroblewski *et al.* (1988).

Our analysis leads to three distinct thresholds R_0 , R_1 and R_2 which are expressible in terms of the model parameters, and which determine the

existence and stability of various states of this system. In particular, we shall prove that if $R_0 < 1$ the plankton populations cannot be sustained and they die out. If $R_0 > 1$ but $R_1 < 1$, the phytoplankton population has a stable steady-state while the zooplankton population cannot be sustained and it dies out. When $R_0 > 1$, $R > 1$ and $R_2 < 1$, there is a coexisting stable steady-state with both phytoplankton and zooplankton populations converging to it. Finally, when R_0 , R_1 and R_2 are larger than one, no stable steady-states exist, and the plankton populations tend to a periodic oscillatory coexisting state (see Figs 2–5). This oscillatory state of the system was overlooked by Wroblewski *et al.* (1988), and its possible significance will be discussed later.

Our treatment of this model leads us to explicit expressions for the steady-states and for the threshold parameters R_0 , R_1 and R_2 . This allows us to study in detail the effects of variations of the pertinent parameters on the stability of the system. The biological interpretation of these results will be presented in the sections that follow.

The remainder of this paper is organized as follows: Section 2 contains the governing dynamic equations of our model and our formal mathematical results which form the basis for all of our conclusions. The final Section (3) is devoted to the discussion of our results and their biological interpretations. Particular attention is paid to the parameter ranges given by Wroblewski *et al.* and to the implications of our results in that situation.

2. Stability and Dynamics of the Plankton System. The system consisting of Phytoplankton (P), Zooplankton (Z) and nitrogen source, say Nitrates (N) satisfies the following set of different equations:

$$\left. \begin{aligned} \frac{dP}{dt} &= aP \frac{N}{k+N} - b(1 - e^{-cP})Z - rP \\ \frac{dZ}{dt} &= Z[\delta(1 - e^{-cP}) - \varepsilon] \\ \frac{dN}{dt} &= -aP \frac{N}{k+N} + rP + \varepsilon Z + \delta_1(1 - e^{-cP})Z. \end{aligned} \right\} \quad (3)$$

In this set, all parameters are non-negative and $\delta + \delta_1 = b$. That is, the phytoplankton which is removed through zooplankton predation ends up either in adding to the zooplankton biomass or to the available nitrate. We note that this is an assumption which is valid only during parts of the season when phytoplankton loss due to cell sinking is negligible.

Adding the equations in (3) we note that $d(P + N + Z)/dt = 0$, thus:

$$P(t) + N(t) + Z(t) = P(0) + N(0) + Z(0) \equiv N_T,$$

where N_T is a constant determined by the initial conditions. Because of this balance, equations (3) can be rewritten as a system involving only P and Z :

$$\left. \begin{aligned} \frac{dP}{dt} &= aP \frac{N_T - P - Z}{k + N_T - P - Z} - b(1 - e^{-cP})Z - rP \\ \frac{dZ}{dt} &= Z[\delta(1 - e^{-cP}) - \varepsilon], \end{aligned} \right\} \quad (4)$$

with $N_T - P(t) - Z(t) = N(t) \geq 0$. Hence, the study of (4) yields complete information on the variables in the system (3). We will treat only the biologically feasible solutions with $P, Z, N \geq 0$.

The dynamics of the zooplankton system (3) are detailed in the following theorem whose proof is the main objective of this section. There are three pertinent natural thresholds R_0 , R_1 and R_2 which are defined in the statement of our theorem and which govern the existence and stability of different states of the system.

Main theorem. The system (3) always has a plankton-free steady-state: $(P, Z, N) = (0, 0, N_T)$. This steady-state is stable if, and only if:

$$R_0 = \frac{aN_T}{r(k + N_T)} < 1, \quad (5)$$

in which case it attracts all feasible solutions. When $R_0 > 1$, the plankton-free steady-state is unstable and there exists a feasible zooplankton-free steady-state $(\bar{P}, 0, \bar{N})$ with:

$$\bar{P} = N_T - \frac{rk}{a-r} = \frac{r(k + N_T)}{a-r} [R_0 - 1], \quad \bar{N} = \frac{rk}{a-r}. \quad (6)$$

This steady-state is stable if and only if $R_0 > 1$ and:

$$R_1 = \frac{\delta(1 - e^{-c\bar{P}})}{\varepsilon} < 1, \quad (7)$$

in which case it attracts all feasible solutions except those with $P(t) = P(0) = 0$. When $R_1 > 1$, the zooplankton-free steady-state is unstable, and there exists a third steady-state (P^*, Z^*, N^*) with the phytoplankton and zooplankton coexisting in equilibrium, and with:

$$P^* = -\frac{1}{c} \ln\left(1 - \frac{\varepsilon}{\delta}\right), \quad N^* = N_T - P^* - Z^*, \quad (8)$$

and Z^* being the unique solution, satisfying $0 < Z^* < N_T - P^*$, of the quadratic:

$$Z^2 - Z \left[\frac{\delta(a-r)}{b\varepsilon} P^* + k + N_T - P^* \right] + P^* \frac{\delta(a-r)}{b\varepsilon} \left[N_T - \frac{kr}{a-r} - P^* \right] = 0. \quad (9)$$

This coexisting steady-state is stable, if and only if $R_0 > 1$, $R_1 > 1$ and $R_2 < 1$, where:

$$\frac{1}{R_2} = \left(bcZ^*(1 - \varepsilon/\delta) + \frac{ak(k + N_T - Z^*)}{k + N_T - Z^* - P^*} \right) / (a - r), \quad (10)$$

in which case this steady-state attracts all feasible solutions which start sufficiently close to it. Moreover, $P^* < \bar{P}$ whenever P^* is feasible. Finally, when $R_2 > 1$, the coexisting steady-state is unstable, and a new oscillatory, coexisting solution exists, which is periodic in time. Moreover, in this case, all feasible solutions other than those with $P(t) \equiv 0$ or $Z(t) \equiv 0$, and (P^*, Z^*, N^*) , tend to a periodic oscillatory state.

This result describes the complete dynamics of the plankton system (3) except for the details of the periodic solutions when $R_2 > 1$. As we shall show later, some details can be given even for the oscillatory states in certain restricted parameter ranges. The results collected in our main theorem can be represented in two diagrams (often called bifurcation diagrams) which can be made quite simple if we notice that:

$$R_1 < 1 \text{ if } \delta < \varepsilon.$$

Also we define the bifurcation parameter λ as follows (Fig. 1):

$$\lambda = \begin{cases} R_0 & \text{if } R_0 \leq 1 \\ 1 + R_1 & \text{if } R_0 \geq 1 \text{ and } R_1 \leq 1 \\ 2 + R_2 & \text{if } R_0 \geq 1, R_1 \geq 1. \end{cases}$$

The proof of our main theorem will be given *via* a series of lemmas which will also provide more detailed information on the dynamic behaviour of the plankton system (3). Our first lemma addresses the existence of the three different steady-states. Since $N(t) = N_T - P(t) - Z(t)$ holds, we will deal with system (4) for P and Z only.

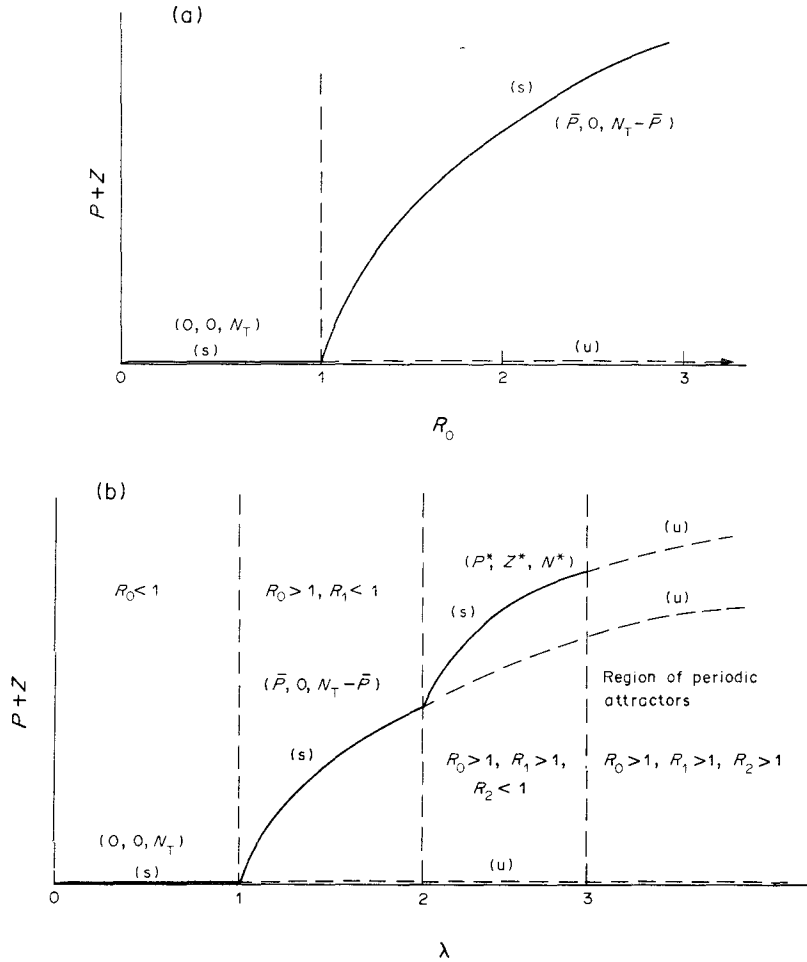


Figure 1. Bifurcation diagram for the plankton model. Solid lines denote stable steady-states (s) and dashed lines denote unstable steady-states (u): (a) the case where $\delta \leq \varepsilon$; (b) the case where $\delta > \varepsilon$.

LEMMA 1. *The system (4) leaves the feasibility region:*

$$R = \{(P, Z): P \geq 0, Z \geq 0, P + Z \leq N_T\},$$

invariant. The system has the following feasible steady states: (a) the plankton free steady state $(0, 0)$ always exists and is the unique steady-state of the system when $R_0 \leq 1$; (b) the zooplankton free steady-state $(\bar{P}, 0)$ exists if, and only if, $R_0 > 1$, and it is the unique non-zero steady-state when $R_1 \leq 1$; (c) the coexisting steady-state (P^, Z^*) exists if, and only if, $R_1 > 1$, and it is the unique steady-state inside the feasibility region R . Moreover, when P^* exists we have $P^* < \bar{P}$.*

Proof. The invariance of R is obtained by noting that for (4):

$$\left. \frac{dP}{dt} \right|_{P=0} = 0, \left. \frac{dZ}{dt} \right|_{Z=0} = 0 \quad \text{and} \quad \left(\frac{dP}{dt} + \frac{dZ}{dt} \right) \Big|_{P+Z=N_T} < 0,$$

since $b > \delta$. Also, it is clear that the right hand side of (4) vanishes when $P = Z = 0$, thus $(0, 0)$ is always a steady-state and is the only steady-state with $P = 0$. Next, any steady-state with $P > 0$ must satisfy either:

$$Z = 0 \quad \text{and} \quad a \left(\frac{N_T - P}{k + N_T - P} \right) - r = 0, \quad (11)$$

or:

$$Z > 0, \delta(1 - e^{-cP}) - \varepsilon = 0 \quad \text{and} \\ aP \left(\frac{N_T - P - Z}{k + N_T - P - Z} \right) - b(1 - e^{-cP})Z - rP = 0. \quad (12)$$

Since feasible solutions must be in R , it is clear that (11) has no feasible solution $P \in (0, N_T)$ if $R_0 \leq 1$. Also, the left hand side of (12) is less than or equal to $rP[R_0 - 1] - b(1 - e^{-cP})Z$, which is negative when $P \in [0, N_T]$, $Z > 0$ and $R_0 \leq 1$, thus (12) has no feasible solutions in this case. This establishes part (a) of Lemma 1.

Next, if $R_0 > 1$, (11) can be easily solved to yield the zooplankton-free solution $(\bar{P}, 0)$ given by (6). Noting that $R_0 > 1$ implies that $a > r$, it follows immediately from (6) that this solution is feasible. Since (11) has no other solutions, $(\bar{P}, 0)$ will be the unique non-zero solution (when $R_0 > 1$) provided that (12) has no feasible solutions. Hence, the proof of (b) will be completed once we have established (c).

In order to prove (c), we first note from (12) that we have $P = P^*$ as given by (8) and that $0 < P^* \leq N_T$, if and only if:

$$0 < -\frac{1}{c} \ln(1 - \varepsilon/\delta) \leq N_T,$$

that is, $0 < \varepsilon/\delta \leq 1 - e^{-cN_T}$. From the second equation (12) we note that Z^* must satisfy (9). Now since $a > r(R_0 > 1)$, the sum of the roots of this quadratic is:

$$\frac{\delta(a-r)}{b\varepsilon} P^* + k + N_T - P^* > 0,$$

and the product of these roots is:

$$\frac{\delta(a-r)}{b\varepsilon} P^* \left[N_T - \frac{kr}{a-r} - P^* \right] \leq 0 \quad \text{when} \quad N_T - \frac{kr}{a-r} - P^* \leq 0.$$

That is when $\bar{P} \leq P^*$ the product of roots is non-positive. However, in this case, one of the roots of (9) is not positive, hence $Z^* \geq$ sum of the two roots. That is:

$$P^* + Z^* > \frac{\delta(a-r)}{b\varepsilon} P^* + k + N_T > N_T,$$

and (P^*, Z^*) is not a feasible solution. Hence, $\bar{P} > P^*$ is necessary for (P^*, Z^*) to be feasible. However $\bar{P} > P^*$ is equivalent to $R_1 > 1$, and it remains to show that this condition is also sufficient for (P^*, Z^*) to be feasible. Thus, assuming that $R_1 > 1$, by the above discussion, we have that both roots of (9) have positive real parts. We next show that when $R_1 > 1$ both of these roots are real and only one of them yields a feasible equilibrium (P^*, Z^*) .

First, from Lemmas 2 and 3, $(0, 0)$ and $(\bar{P}, 0)$ are saddle points with stable manifolds $P=0$ and $Z=0$, respectively, whenever $R_0 > 1$ and $R_1 > 1$. Thus solutions of (4) with initial data inside R cannot tend to either of these steady-states as $t \rightarrow \infty$. Since R is invariant, by the Poincaré–Bendixson theorem, these solutions must tend to either a steady-state inside R or to a limit cycle inside R . However, every limit cycle must contain a steady-state, thus there must exist a steady-state inside R . However, any such steady-state must be a solution of (12). Thus the roots of (9) must both be real and positive in this case, and at least one of these must yield a feasible state (P^*, Z^*) .

The comparison of (P^*, Z^*) for $R_0 > 1$, $R_1 > 1$ proceeds as follows. In Lemmas 2 and 3 we show that the linearizations of the right hand side of (4) about $(0, 0)$ and $(\bar{P}, 0)$ have no zero eigenvalues. When $R_0 > 1$ and $R_1 = 1$, the linearization about $(\bar{P}, 0)$ has a single zero eigenvalue. By the extended implicit function theorem, there is a unique branch that bifurcates from $(\bar{P}, 0)$, and as we showed above, enters the interior of R . Since no other bifurcations occur at $(0, 0)$ and $(\bar{P}, 0)$ and there are no equilibria at any other points of the boundary of R , there is at most one equilibrium inside R wherever $R_0 > 1$, and $R_1 > 1$. This completes the proof of part (c) and of Lemma 1. ■

Remark. An alternative proof of the uniqueness of (P^*, Z^*) can be obtained by using a homotopy argument on the roots of (9) as functions of the parameters of that equation. One first notes that for $R_0 > 1$ we have $0 < \bar{P} < N_T$ and when $R_1 = 1$, $P^* = \bar{P}$, hence (9) has the roots $Z_1^* = 0$, $Z_2^* = \bar{P}(\delta(a-r)/b\varepsilon) + k + N_T - \bar{P}$. The second root is not feasible since $Z_2^* + P^* + Z_2^* = \bar{P} > N_T$. Now, as R_1 increases, the root Z_1^* becomes positive by the proof of our lemma, while Z_2^* can become feasible only if $Z_2^* + P^* = N_T$ at some value of $R_1 > 1$. But, as we have shown already, the sum of the right hand sides of (4) is strictly negative

when $P+Z=N_T$ and $R_0>1$. Thus $Z_2^*+P^*>N_T$ for all $R_1>1$ and there is at most one feasible solution of (12).

In the next three lemmas we established our results on the stability of the feasible equilibria. We also provide some additional details on the global attractivity of these equilibria.

LEMMA 2. *The plankton-free steady-state $(0, 0)$ is locally asymptotically stable if and only if $R_0<1$. When it is stable it attracts all the feasible solutions. When $R_0>1$, $(0, 0)$ is a saddle point with stable manifold $P=0$.*

Proof. Linearizing (4) about $(0, 0)$ we obtain a system whose Jacobian matrix is:

$$J(0, 0) = \begin{bmatrix} r(R_0-1) & 0 \\ 0 & -\varepsilon \end{bmatrix},$$

which is stable if and only if $R_0<1$. By Lemma 1, when $R_0<1$ $(0, 0)$ is the only steady-state in the invariant region R , hence, by the Poincaré-Bendixson Theorem it must attract all solutions in R . Finally, when $R_0>1$, the eigenvalues of $J(0, 0)$ are $r(1-R_0)>0$ and $-\varepsilon$, hence it is a saddle. From the diagonal form of $J(0, 0)$ it is clear that its stable manifold is $(0, 1)$, i.e. the axis $P=0$. Finally, from (4) when $P=0$ we have $dZ/dt = -\varepsilon Z$, $dP/dt = 0$, hence the axis $P=0$ is the stable manifold of the non-linear system (4) as well. ■

LEMMA 3. *Suppose $R_0>1$. The zooplankton-free steady-state $(\bar{P}, 0)$ is locally asymptotically stable if and only if $R_1<1$. When it is stable it attracts all feasible solutions except those with initial data $P(0)=0$ (hence $P(t)\equiv 0$). When $R_1>1$, $(\bar{P}, 0)$ is a saddle point with stable manifold $\{(P, Z): Z=0, 0<P\leq N_T\}$.*

Proof. The linearization of (4) about $(\bar{P}, 0)$ yields a system whose Jacobian matrix is:

$$J(\bar{P}, 0) = \begin{bmatrix} J_{11} & J_{12} \\ 0 & J_{22} \end{bmatrix},$$

with:

$$\left. \begin{aligned} J_{11} &= (1-R_0) \frac{r(k+N_T)}{k} < 0, \quad J_{22} = \varepsilon(R_1-1), \\ J_{12} &= -\frac{\varepsilon b}{\delta} R_1 - \frac{(r-a)}{ak} [N_T(r-a) + rk]. \end{aligned} \right\} \quad (13)$$

Since $J(\bar{P}, 0)$ is triangular, its eigenvalues are J_{11} and J_{22} . Clearly it is stable

when $R_1 < 1$ whereas when $R_1 > 1$ it is a saddle point with stable manifold coinciding with the axis $Z=0$. By Lemma 1, when $R_1 < 1$, there are no steady-states inside R and since $(0, 0)$ is a saddle point (recall that $R_0 > 1$), it cannot attract any solutions with initial values in the interior of R . Thus, $(\bar{P}, 0)$ attracts all solutions in R except those lying on the stable manifold of $(0, 0)$, that is, those with $P(0)=0$. Finally, when $Z=0$, $dZ/dt=0$ and:

$$\frac{dP}{dt} = rP \left[\frac{a(N_T - P)}{r(k + N_T - P)} - 1 \right], \quad (14)$$

since $0 \leq P \leq N_T$. The right hand side of (14) vanishes at $P=0$ and $P=\bar{P}$, and its derivative at $P=0$ is $r(R_0 - 1) > 0$ and at $P=\bar{P}$ it is negative. Thus $(\bar{P}, 0)$ attracts all solutions with initial data in $\{(P, Z): Z=0, 0 < P \leq N_T\}$. ■

LEMMA 4. Suppose that $R_0 > 1$ and $R_1 > 1$. The coexisting steady-state (P^*, Z^*) is unstable, going through a stable Hopf bifurcation at $R_2 = 1$. In this case there exists at least one periodic solution surrounding (P^*, Z^*) and all feasible solutions other than (P^*, Z^*) and those with initial data having either $P(0)=0$ or $Z(0)=0$, tend to a periodic oscillatory state.

Proof. The linearization of (4) about (P^*, Z^*) yields a system whose matrix is:

$$J(P^*, Z^*) = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & 0 \end{bmatrix}, \quad (15)$$

with:

$$\left. \begin{aligned} J_{11} &= (a-r) \left(1 - \frac{1}{R_2} \right), \\ J_{12} &= -\frac{b\varepsilon}{\delta} - \frac{akP^*}{(k + N_T - P^* - Z^*)^2} < 0, \\ J_{21} &= c\delta(1 - \varepsilon/\delta)Z^* > 0. \end{aligned} \right\} \quad (16)$$

We immediately note that $\text{Det } J(P^*, Z^*) = -J_{12}J_{21} > 0$ and $\text{Trace } J(P^*, Z^*) = J_{11} = (a-r)(1 - 1/R_2)$. Thus, (P^*, Z^*) is locally asymptotically stable if, and only if, $R_2 < 1$ (recall that $R_0 > 1$, hence $(a-r) > 0$). In this case, by Lemmas 1–3, the steady-states $(0, 0)$ and $(\bar{P}, 0)$ do not attract any solutions with initial data in the interior of R . Next, from (16):

$$\frac{\partial}{\partial R_2} J_{11} = \frac{\partial}{\partial R_2} \text{Trace } J(P^*, Z^*) > 0 \text{ at } (P^*, Z^*).$$

Hence, when R_2 increases past $R_2=1$, $J(P^*, Z^*)$ has a pair of complex conjugate eigenvalues which cross the imaginary axis transversely. Hence, there is a Hopf bifurcation when (P^*, Z^*) loses its stability at $R_2=1$, and the periodic solution which bifurcates from (P^*, Z^*) has period of mean $2\pi/\omega$, where:

$$\omega = \sqrt{J_{11}J_{21}}. \quad (17)$$

In this J_{11} and J_{21} are given by (16). The stability of this Hopf bifurcation can be determined by using the criterion in (A16) of Liu *et al.* (1986). Computing the expression given in (A16) of that reference, we note that it is negative at $R_2=1$, and hence, the bifurcating periodic solution is locally asymptotically stable when $R_2-1>0$ is small. Finally when $R_2>1$ all three steady states in R are unstable and by our Lemmas there are no solutions which start inside R and tend to $(0, 0)$ or $(\bar{P}, 0)$. Hence, there are no phase polygons or heteroclinic limit cycles in R . By the Poincaré-Bendixson Theorem, all solutions starting in R must tend to some periodic solution. ■

Remark 1. The method in Liu *et al.* (1986) which we used to settle the local stability of the bifurcating periodic solution says that, if the expression given in (A16) of the reference is negative at $R_2=1$, then the Hopf bifurcation is supercritical. That is, the bifurcated periodic solutions occur only to the right of the point 3 on the λ -axis in Fig. 1b.

Remark 2. We conjecture that the periodic solution which bifurcates from (P^*, Z^*) is unique and is the global attractor inside R when $R_0>1$, $R_1>1$ and $R_2>1$. We have not yet been able to prove this, but our numerical work strongly supports the conjecture. Further work is in progress on this point. The results quoted in Remark 1 shows that the periodic solutions are stable near $R_2=1$.

Remark 3. $P^* = -(1/c) \ln(1-\varepsilon/\delta) \rightarrow 0$ as either $c \rightarrow \infty$ or $\delta \rightarrow \infty$, or else $\varepsilon \rightarrow 0$. All three of these limiting cases represent an extreme in the efficiency of the zooplankton utilization of its phytoplankton prey. Also, if $\tilde{P}$ is a periodic attractor (R_2 must be >1 for $\tilde{P}$ to exist), then it is easy to see from the second equation in (3) that the average value of $e^{-c\tilde{P}}$ over a period T is given by:

$$\frac{1}{T} \int_0^T e^{-c\tilde{P}(t)} dt = \frac{\delta - \varepsilon}{\delta}. \quad (18)$$

This tends to 1 as $\delta \rightarrow \infty$, hence $\tilde{P}(t)$ also tends to zero as $\delta \rightarrow \infty$. This is seen by noting from (3) that:

$$\left| \frac{d\tilde{P}(t)}{dt} \right| \leq |a - r|\tilde{P}(t),$$

and by using (18).

COROLLARY. *When $R_0 > 1$, $R_1 > 1$, and $R_2 > 1$, all limiting periodic solutions $(\tilde{P}(t), \tilde{Z}(t), \tilde{N}(t))$ obey:*

$$\frac{1}{T} \int_0^T \tilde{P}(t) dt > P^*,$$

where T is the period of the solution. Thus the average value of the phytoplankton exceeds the coexisting steady-state value.

Proof. From the second equation in (3) we have:

$$0 = \int_0^T \frac{1}{\tilde{Z}} \frac{d\tilde{Z}}{dt} dt = \delta \int_0^T (1 - e^{-c\tilde{P}}) dt - \varepsilon T,$$

since $\tilde{Z}(t)$ is periodic with period T . Thus:

$$\left(1 - \frac{\varepsilon}{\delta}\right) = \frac{1}{T} \int_0^T e^{-c\tilde{P}} dt > e^{-1/T \int_0^T \tilde{P} dt}.$$

This follows by the concavity of the function $x \rightarrow e^{-cx}$, the fact that $\tilde{P}(t) > 0$ and the fact that:

$$\frac{1}{T} \int_0^T \tilde{P} dt > -\frac{1}{c} \ln(1 - \varepsilon/\delta) = P^*. \quad \blacksquare$$

This corollary shows that the phytoplankton has a higher average population in its coexisting oscillatory state than at the steady coexisting state. Hence, on the average it has an advantage in the oscillatory coexisting state, even though its population levels need to drop below P^* for parts of the oscillatory state.

3. Discussion and Conclusion. The following table sets out the results that we have established for the three component phytoplankton trophic chain model.

There are two gaps in this table. The first occurs when $R_0 > 1$, $R_1 > 1$, $R_2 < 1$ where we have shown that the coexisting equilibrium is locally asymptotically stable (hence attracts all states which start near it) but we have yet to show that

Table 1. Thresholds and stability of steady-states

Thresholds R_0, R_1, R_2	Steady-state properties (0, 0, N_T) ($\bar{P}, 0, N_T - \bar{P}$) (P^*, Z, N^*)			Periodic attractors
$R_0 < 1$	Stable and global attractors	Not feasible	Not feasible	Not feasible
$R_0 > 1, R_1 < 1$		Stable and global attractor	Not feasible	Not feasible
$R_0 > 1, R_1 > 1, R_2 < 1$	Unstable	Unstable	Stable	—
$R_0 > 1, R_1 > 1, R_2 > 1$	Unstable	Unstable	Unstable	Attracts solutions

it is the global attractor. This can be settled by showing that no periodic solutions exist except in this case. We do know that there is no Hopf bifurcation from $R_2 = 1$ extending into $R_2 < 1$. The other gap in this table which occurs when $R_0 > 1, R_1 > 1$ and $R_2 > 1$, where we have not demonstrated that there cannot be more than one periodic solution. We have proved that every state which starts off with some zooplankton and phytoplankton present must tend to a period oscillatory coexisting state. It is the uniqueness of this state that remains an unsolved question. However, we know several properties of the periodic coexisting state, some of which have important biological implications.

The thresholds R_0, R_1 and R_2 have relevant biological interpretations. Firstly:

$$R_0 = \frac{1}{r} \frac{aN_T}{k + N_T},$$

is the ratio of the maximal nutrient uptake of the phytoplankton to its death rate in the absence of any zooplankton. It measures the ability of the nutrient environment to support a phytoplankton population and can be understood through the following reasoning. If in a plankton-free layer with nutrient level N_T one introduces a single phytoplankton at time $t=0$, then it will have probability e^{-rt} of surviving to time $t>0$, and the expected number of its offspring, over its life-time would be:

$$\int_0^\infty e^{-rt} \left(\frac{aN_T}{k + N_T} \right) dt = \frac{1}{r} \frac{aN_T}{k + N_T} = R_0. \quad (19)$$

Thus, if $R_0 < 1$, there would be, on average, less than one offspring over the lifetime of each phytoplankton and the nutrient level would be insufficient to support a stable phytoplankton population. If $R_0 > 1$, then the average reproduction of each phytoplankton is more than one and a stable population can be established. Note that, as would be expected, R_0 is an increasing

function of the total nutrient level N_T and of the uptake rate a , and a decreasing function of the death rate r and the half saturation constant k . We also note that the phytoplankton equilibrium level:

$$\bar{P} = N_T - \frac{rk}{a-r}$$

(here $a > r$) increases with N_T and a and decreases with r and k .

The second threshold:

$$R_1 = \frac{\delta(1 - e^{-c\bar{P}})}{\varepsilon},$$

is the ratio of the maximal uptake rate of the zooplankton to its death rate, given that a stable phytoplankton population $\bar{P}$ has been established. It can be interpreted in exactly the same way as R_0 was in the previous paragraph by considering the introduction of a single zooplankton in an established phytoplankton population which has settled to its equilibrium level $\bar{P}$. R_1 is clearly an increasing function of the uptake rate δ , the saturation exponent c , and the phytoplankton equilibrium level $\bar{P}$ (hence, also of the total nutrient level N_T). As expected, it is a decreasing function of the zooplankton death rate ε . Our results show that a zooplankton population can establish itself in coexistence with a phytoplankton population if and only if $R_1 > 1$, that is, the expected lifetime productivity of each zooplankton is greater than one.

The final threshold R_2 is more complicated, and reflects certain aspects of the "enrichment paradox". When $R_2 < 1$, the phytoplankton population settles to the coexisting steady-state level P^* if any zooplankton is present, and to the level $\bar{P}$ if no zooplankton at all is present. Since we have proved that $P^* < \bar{P}$, the phytoplankton population levels decrease to P^* when N_T increases to the point where $R_1 > 1$ from their level at $\bar{P}$ when $R_1 < 1$. Thus enriching the environment decreases the steady-state phytoplankton population level. A further enrichment of the environment which drives R_2 beyond its threshold level of 1, destabilizes the coexistence steady state (P^*, Z^*, N^*) . Thus enrichment in this case has a destabilizing effect, leading to a periodic oscillatory coexistence state. As we have shown, however, in this oscillatory state, the phytoplankton attains for part of the time population levels which exceed the zooplankton-free steady-state level $\bar{P}$. Thus, environmental enrichment allows increases in population levels in this case, but only in an oscillatory state. In Figs 2 and 3 we have shown the dependence of R_1 and R_2 on the parameters N_T , ε and $kr/(a-r)$ and indicate the parameter regions where the various types of stable behaviours occur.

The type of threshold parameter combinations R_0 , R_1 and R_2 that we have used here are similar to those that occur in the theory of epidemics. The reader

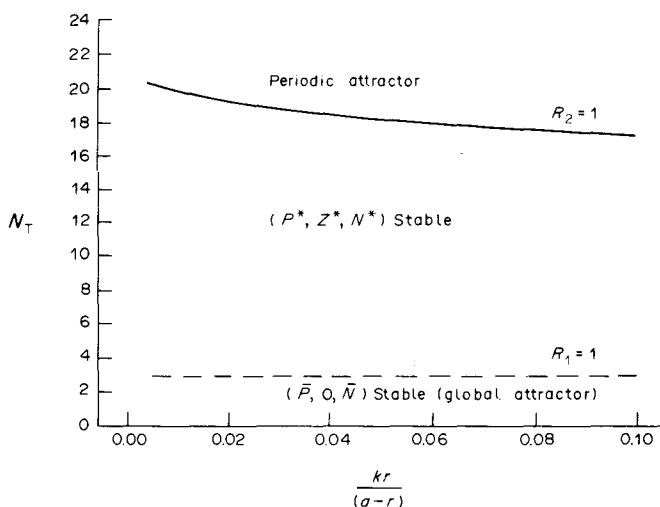


Figure 2. Stability regions of steady-states for $R_0 > 1$ and various values of N_T and $kr/(a-r)$. Solid line denotes $R_2 = 1$ and dashed line denotes $R_1 = 1$.

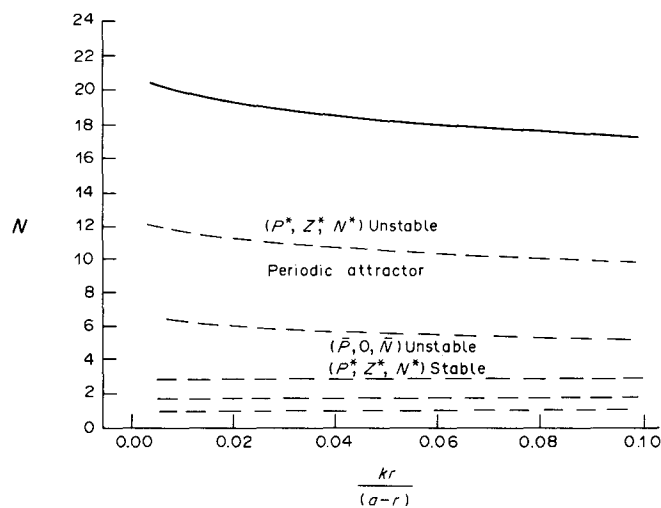


Figure 3. Stability regions of steady-states for various values of N_T , $kr/(a-r)$ and ϵ : — corresponds to the true period obtained by integrating the system of differential equations (3) (R_2 is much greater than 1.); - - - corresponds to the period $2\pi/\omega$ obtained from the Hopf bifurcation formula (equations 17 and 16). (Here R_2 is greater but very close to 1.); - · - corresponds to R_2 (equation 10) for which equation (3) is integrated to obtain the true period.

is referred to Busenberg and Von der Driessche (1989) for a detailed discussion in that setting.

The values used by Wroblewski *et al.* (1988) are given in Table 2. In order to get different solutions we varied the depth h and total nutrients N_T .

Table 2. Parameter values used by Wroblewski *et al.* (1988) and for all the computations in this paper

Parameter	Value	Dimensions
a	2.0	day^{-1}
k	0.2	$0.1 \mu\text{g atom NO}_3/\text{dm}^3$
r	0.1	day^{-1}
b	0.5	day^{-1}
δ	0.35	day^{-1}
ε	0.2	day^{-1}
δ_1	0.15	day^{-1}
c	0.5	$(\mu\text{g atom NO}_3/\text{dm}^3)^{-1}$

The destabilization of the coexistence equilibrium (P^* , Z^* , N^*) when $R_2 > 1$ has important implications in the use of this type of trophic level model to interpret data from lakes and from oceans as in Wroblewski *et al.* (1988). In ocean situations as well as in layered lakes where there is periodic mixing of the trophic layers due to climatic conditions, the stable steady-states and the periodic attractors of this system are significant only when the time required to settle to them is short compared to the average time interval between two mixing events. These settling times can be estimated from our analysis once the various parameter values pertinent to the situation have been estimated. The settling time for specific parameter values can also be obtained by numerically integrating the differential equation system (4). In Fig. 4 we show the results of such numerical integrations, including a situation where $R_0 > 1$, $R_1 > 1$ and $R_2 > 1$, where there is a periodic oscillatory coexisting state.

We note that if we use the parameter values reported by Wroblewski *et al.* (1988), certain regions of the North Atlantic during the month of May do lead to situations where the steady-states are unstable and the plankton and nutrient levels tend to an oscillatory state. Thus the analysis of the data in that paper, which is based only on the determination of steady-states, is not valid in these situations, and a time average of the periodic states needs to be used instead of the values of P^* at the steady-state. It is of interest to determine whether observed fluctuations of the phytoplankton levels are due to these stable oscillations rather than to random effects, as is often supposed. The period of these oscillations near their point of bifurcation (i.e. for $R_2 > 1$ but with R_2 near to 1) is explicitly given by (16) and (17). When R_2 is much larger than 1, the period can be computed numerically by integrating the differential

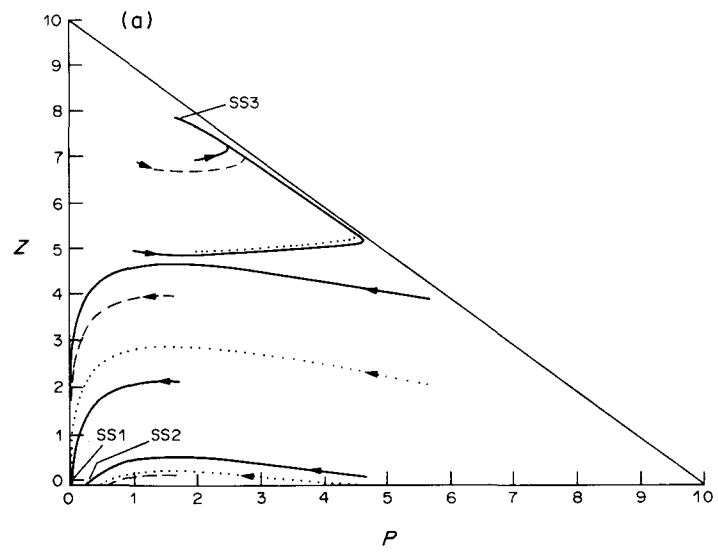


Figure 4(a).

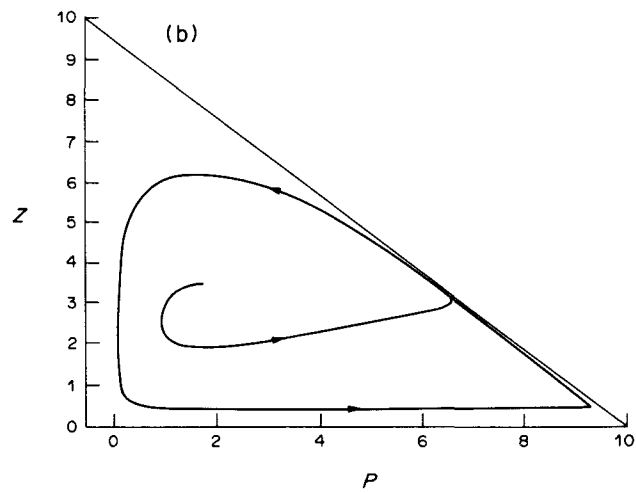


Figure 4(b).

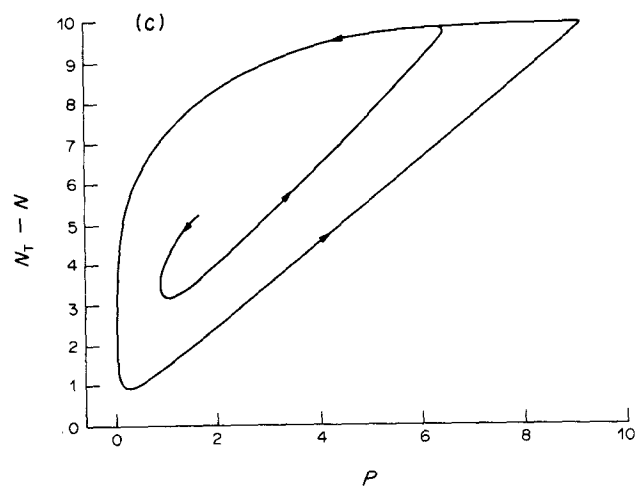


Figure 4(c).

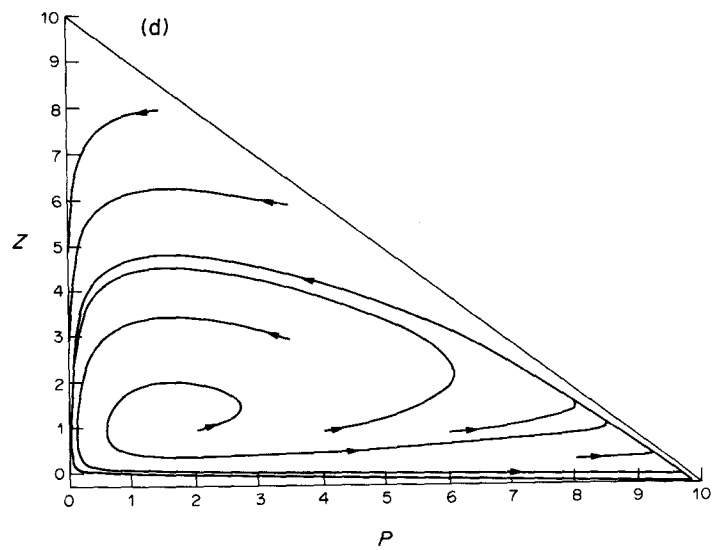


Figure 4(d).

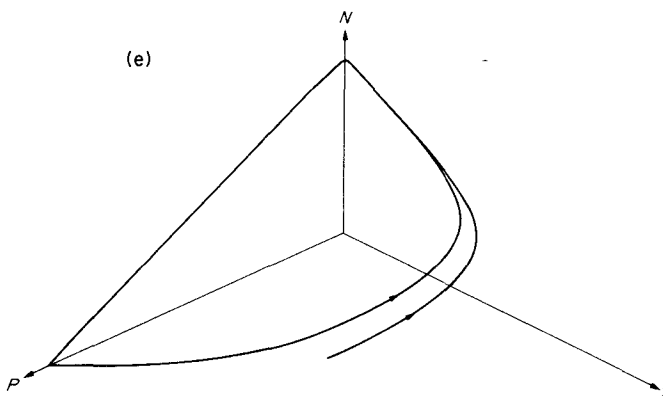


Figure 4(e).

Figure 4. (a) Projection of the trajectories on the PZ plane for various values of R_0 , R_1 and R_2 . All the trajectories approach the steady-state $SS1 = (0, 0, N_T)$ for $R_0 < 1$. ($R_0 = 0.31$.) All the trajectories approach the steady state $SS2 = (\bar{P}, 0, \bar{N})$ for $R_0 > 1$ and $R_1 < 1$. ($R_0 = 1.004$; $R_1 = 0.996$.) All the trajectories approach the steady-state $SS3 = (P^*, Z^*, N^*)$ for $R_0 > 1$, $R_1 > 1$ and $R_2 < 1$. ($R_0 = 1.009$; $R_1 = 1.374$; $R_2 = 0.993$.) Orbit of the periodic solution for $R_0 > 1$, $R_1 > 1$ and $R_2 > 1$. ($R_0 = 6.16$; $R_1 = 1.74$; $R_2 = 2.65$.) (b) P vs Z ; (c) P vs $N_T - N$. (d) Orbit of the periodic solution in the PZ-plane for $R_0 > 1$, $R_1 > 1$ and $R_2 > 1$. ($R_0 = 6.16$; $R_1 = 1.74$; $R_2 = 2.65$.) (e) A three-dimensional view of the phase plane showing the approach to and the limit cycle in the PNZ space. This view is normal to the phase plane. ($R_0 > 1$; $R_1 > 1$ and $R_2 > 1$). ($R_0 = 6.16$; $R_1 = 1.74$; $R_2 = 2.65$.)

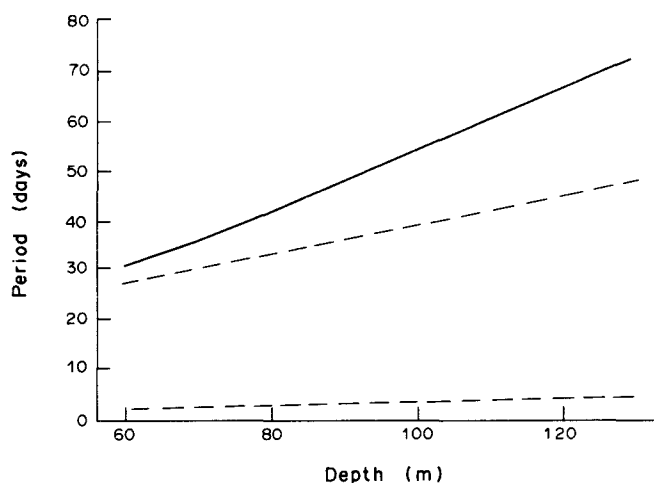


Figure 5. Period and threshold (R_2) vs the layer depth (h) for $N_T = 10.0$. — corresponds to the true period obtained by integrating the system of differential equations (3). R_2 is much greater than 1. --- corresponds to the period $2\pi/\omega$ obtained from the Hopf bifurcation formula (equations (17) and (16)). Here R_2 is greater but very close to 1. — — — corresponds to R_2 (equation (10)) for which equation (3) is integrated to obtain the true period.

equation (3). We have done these computations for the parameter ranges given in Wroblewski *et al.* (1988) and show them in Fig. 5, below. In converting the information on layer depth h to the maximal nutrient uptake parameter of our model we have used the relation:

$$a = \frac{20}{h} \left(\frac{1 - e^{0.1h}}{1 - e^{-1}} \right),$$

which corresponds to the data and formula given in that reference.

It is clear from this analysis of the data of Wroblewski *et al.* (1988) that the initial periods at the bifurcation point vary from 8.7 to 38.7 days. This full range of periodic behaviour could be expected to occur in the above parameter ranges and could lead to fluctuations of the plankton levels which at first sight may appear to be random.

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